

Types of Research Data

Broadly categorized as:

- **Primary data:** directly collected by the researcher.
 - Also called raw data.
 - Can be collected through interview, survey, experiment, etc.
 - Observation, record, experiment etc. are the actual activity.
 - Collected for the first time.
 - Expensive.
 - **More reliable.**
 - Example: ?
- **Secondary data:** already collected by someone else.
 - **Derived** from the primary data.
 - Economical.
 - **Less reliable.**
 - Example: ?

Scope of Survey Based Research

- 
- A **quantitative or qualitative research** that is **based on survey** is referred to as survey based research.
 - Survey is a data collection tool to conduct survey based research.
 - Most of the survey based research is non-experimental.
 - Data collection are done from a specific group of people (respondents) to gain information about a topic of interest.
 - Data collection in a survey can be **verbal (interview) or written (questionnaire).**
 - Interview can be taken face-to-face or through telephone.
 - In a face-to-face interview, respondents usually have trust on the **surveyor(s)** and provide honest **opinion/feedback.**
 - A survey through telephone requires less effort and manpower than face-to-face survey.

Survey Based Research

- Survey can be done online or offline.
- Online survey is the most cost-effective way and it is possible to reach the maximum number of people (respondents). But a respondent needs to have the ability to operate the computer and the internet.
- The offline pen and paper based (questionnaire) is usually applied in the field research.

Survey Based Research

- **Advantages of Survey Method:**
 - Surveys can be conducted faster and cheaper compared to other methods of primary data collection such as observation and experiments.
 - Primary data gathered through surveys are relatively easy to analyze.
- **Disadvantages of Survey Method:**
 - In some cases, unwillingness or inability of respondents to provide information.
 - Human bias of respondents, i.e. respondents providing inaccurate information.
 - Differences in understanding: it is difficult to formulate questions in such a way that it will mean exactly the same thing to each respondent.

Process of Implementing Survey Based Research

- **Decide the survey questions:** Survey questions should be set carefully. Questions should be grammatically and logically appropriate. Based on the survey objectives, questions could be open or close ended.
- **Finalize the target participants (respondents):** It is very important to choose the right group of people for conducting a survey.
- **Send out surveys via decided medium:** At this stage, the surveyor(s) send the questions to the respondents and waits patiently to get back their opinion/feedback.
- **Analyze the survey results:** Survey results should be analyzed carefully and appropriately. Patterns need to be identified.

Survey Based Research

- **Sample Size:**
 - Sample is a subset of a target population chosen in order to conduct a survey.
 - It is called a sample because it represents only a part of group of people.
 - Selection of a sample size: (see scanned course material).

Survey Based Research

- Example:

- **Title:** “A Survey on Programming Language Preferences and Their Impact on a Developer’s Productivity”.
- **Objective:** To understand which (and why) programming languages are mostly preferred and how does that affect a developer’s promotion, productivity and working environment.
- **Methodology:** Online based questionnaire. Finalize respondents.
- **Survey Questions:**
- **Analysis:** Statistical tools.
- **Result.**
- **Findings.**
- **Suggestion.**
- **Future Enhancement.**